Nineteen years of ASMR on YouTube: A multilingual, theme-level analysis of 89,241 videos

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Abstract

Autonomous sensory meridian response (ASMR) videos are commonly framed around whispered speech, gentle sounds, close-up attention, relaxation, sleep, and comfort, yet longitudinal, large-scale descriptions of this labelled YouTube genre are scarce. We analyse 89,241 ASMR YouTube videos uploaded between 1 January 2008 and 31 July 2026, with metadata processed on 1 August 2026, from 18,024 channels across 81 language categories. Using title and description text mining with duration, engagement, growth, theme-label indicators, and exploratory clustering metrics, we provide a multilingual longitudinal map of explicitly labelled ASMR on YouTube. Mean growth was 1,125.41 views/day (SD 11,196.23). Among videos with known duration, those under 10 min grew fastest on average (2,075.55 views/day), videos between 10 and 30 min averaged 585.11 views/day, and videos over 180 min averaged 1,293.15 views/day. The most prevalent themes were sleep-related content (17.62%), whisper-focused content (15.49%), role-play (14.09%), mukbang or eating (7.08%), and visual triggers (6.85%); driving-related content accounted for 3.67%. An exploratory k-means solution grouped videos into 11 clusters, and a full-dataset UMAP projection visualised the resulting structure for all 89,241 videos without treating those groups as definitive ASMR subgenres.

1 Introduction

Autonomous sensory meridian response (ASMR) refers to a tingling, soothing sensation that some individuals experience in response to specific audio-visual stimuli such as whispering, gentle tapping, or simulated close personal attention [1, 2]. Viewers often describe ASMR as a form of self-care, reporting reductions in stress and anxiety, improved mood, and better sleep [3, 4]. Such claims are background context for the present study rather than outcomes measured here: our dataset captures video metadata and creator-supplied text, not viewers, subjective states, sleep quality, or physiological responses. Experimental work has increasingly established ASMR as a reproducible psychophysiological phenomenon rather than a purely anecdotal curiosity. Studies such as [1] and [5] document systematic changes in affect, cardiovascular responses, electrodermal activity, and neural dynamics during exposure to ASMR, while narrative and comparative reviews, particularly [6], emphasise heterogeneity in methods, stimuli, and results and situate ASMR in relation to neighbouring constructs such as frisson, synaesthesia, and misophonia.



Fig 1. Word cloud of the 200 most frequent words extracted from the titles of all 89,241 ASMR videos in the dataset after custom stopword removal and text cleaning. Font size is proportional to corpus frequency.

From a media and communication perspective, ASMR is also a socio-cultural practice in which intimacy, care, and affect are mediated through networked screens and headphones. Early qualitative work highlighted the grassroots "whisper community" that formed around online forums and pioneering YouTube channels, showing how whispered speech and low-fi recording practices foster mediated co-presence and "non-standard intimacy". [7] and Smith and [8] analyse how ASMR creators and viewers negotiate pleasure, stigma, and shifting boundaries between public and private life. Later contributions such as [9] and [10] further theorise ASMR as a mode of sonic and audiovisual intimacy linked to broader developments in online video culture.

Industry surveys and platform reports suggest that ASMR or ASMR-like content may be especially important for younger audiences, with trend studies reporting high self-reported use among 16–24-year-olds and Generation Z for relaxation, studying and sleep support [11, 12]. In the peer-reviewed literature, ASMR research has similarly concentrated on young adults–experimental studies often recruit participants in their early 20s–so the strongest empirical evidence currently available mainly reflects younger, online-active cohorts rather than population-representative age gradients [2, 5, 13].

In parallel, a growing body of work asks how video length relates to attention and engagement on digital

platforms. Large-scale analyses of YouTube viewing behaviour show that audience retention generally declines rapidly early in playback and that view-duration trajectories vary substantially across videos, channels, and contexts, implying that there is no single universally optimal length [14]. Academic analyses of YouTube influencers similarly find that medium- and long-form videos tend to attract more views, likes, and comments than very short clips [15]. In short-form environments such as social-media feeds and in-feed advertising, experimental work instead points to inverted-U effects with optimal lengths on the order of a few tens of seconds [16]. Educational video research on Massive Open Online Courses (MOOCs) also documents sharp engagement declines beyond approximately six minutes, while cautioning against a universal "six minute rule" [17]. However, these studies focus on general, promotional or instructional content rather than ASMR, leaving open whether a relaxation-orientated genre follows similar length–engagement patterns or displays its own session-length preferences.

Historically, ASMR on YouTube has developed through overlapping phases. Scholarly accounts frequently point to WhisperingLife's early whisper-only upload *Whisper 1 — hello!* (2009) as a landmark in the emergence of intentionally produced "trigger" content and in consolidating an identifiable whisper/ASMR community around dedicated channels rather than incidental triggers in other genres (`https://www.youtube.com/watch?v=IHtgPbfTgKc`) [18, 19]. In the early origins and niche-community phase (pre-2012), ASMR circulated mainly through small forums and dedicated whisper or soft-spoken role-play channels, following the coining of the term "autonomous sensory meridian response" in 2010 [18, 19]. A subsequent phase of mainstreaming and platform growth (approximately 2012–2018) saw ASMR become increasingly legible as a distinct YouTube genre, as creators and audiences learnt to work with platform features (search, tags, recommendation and monetisation systems) to produce, find, and sustain ASMR content on scale [19, 20]. As the genre matured, a phase of diversification and professionalisation (roughly 2016–2020) introduced recognisable sub-genres (e.g. medical role play, sound-focused "no talking" videos, and eating-trigger content) alongside increasingly sophisticated production practices, including higher-fidelity and spatial audio recording [19, 20]. Most recently, a phase of commercialisation and platform changes (2020–present) has been marked by sponsored and branded ASMR content and creator strategy shifts in response to algorithmic governance and monetisation pressures [20–22].

The growing empirical literature describes different facets of this evolving ASMR ecosystem on YouTube, but it remains dispersed across disciplinary and methodological domains. One strand comprises content-analytic studies that systematically code the visual, auditory, and interactional features of ASMR videos and relate these to viewer engagement. [23], for example, examine interaction modalities and parasocial cues across a large sample of videos, while other studies document trigger types, performer characteristics, and formal conventions in YouTube ASMR content. A second strand focusses on comments and everyday uses of ASMR through netnography and qualitative content analysis. Examples include [24] and [25], which show how viewers frame ASMR as self-care, negotiate authenticity and intimacy, and articulate the meanings of "tingles" in comment threads and online discussions.

A third cluster of work approaches ASMR as a multimodal and discursive phenomenon, often concentrating on role-play sub-genres. Studies such as [26], [19] and [27] combine multimodal discourse analysis with concepts including haptic audio-visuality, ambient co-presence, and digital intimacy to show how camera positioning, gaze, voice, gesture, and sound design are orchestrated to simulate physical closeness and care. Related work on ASMR role play and whispered speech feeds into speech-technology and HCI research: [28] and [29] treat ASMR as a large-scale resource for whispered speech and unvoiced language identification. A fourth strand situates ASMR within broader audience and platform dynamics on YouTube and social media. Studies such as [20, 30] investigate how ASMR creators navigate YouTube's affordances, monetisation regimes, and community norms, while [21] and [22] examine the integration of ASMR into influencer marketing and advertising. More general work on YouTube engagement, such as [31], offers methodological and conceptual resources to understand ASMR as an instance of a wider set of native media practices.

1.1 Aim of the study

The aim of this study is to provide a quantitative multilingual characterisation of ASMR content on YouTube using 89,241 videos uploaded between 2008 and 2026 and retrieved through a large-scale keyword-based

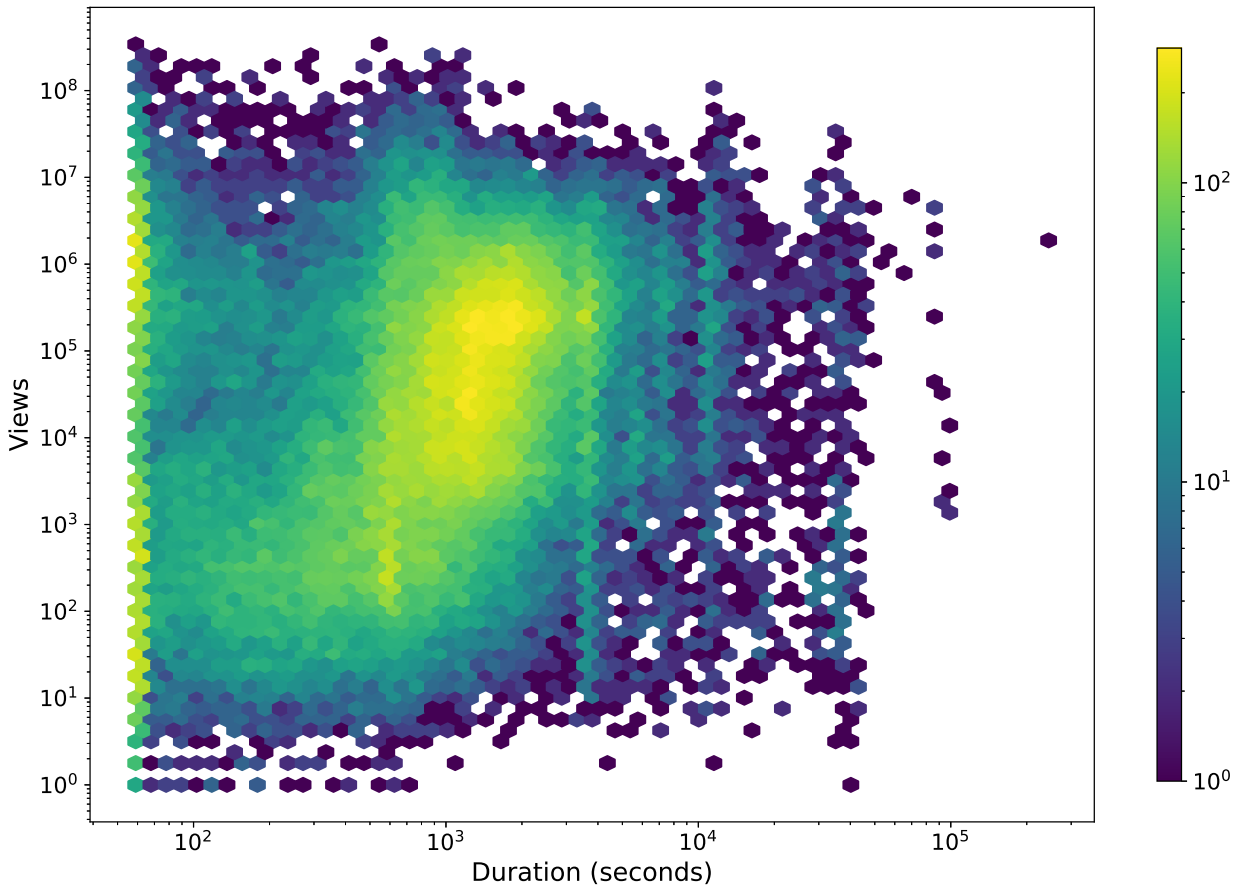


Fig 2. Log–log hexagonal bin plot showing each video's duration in seconds (x-axis) against its cumulative view count at the time of data processing on 1 August 2026 (y-axis). The plot includes 88,482 ASMR videos with positive duration and positive view counts; each hexagon aggregates multiple videos, with colour intensity indicating the number of videos in that bin.

pipeline centred on the query "ASMR". The central contribution is a large-scale multilingual longitudinal map of explicitly labelled ASMR on YouTube, showing how the genre has diversified by language, duration, textual theme labels, and engagement. We combine video-level metadata, language information, title and description text, rule-based theme-label indicators, and behavioural measures such as views per day and mean engagement. Our analysis asks: (i) how ASMR videos are distributed across languages, formats, and title styles; (ii) how prevalent major ASMR textual labels (e.g. whispering, no talking, sleep, binaural, role-play, mukbang, driving) are and how they differ in reach and mean engagement; (iii) how duration and other structural features relate to popularity; (iv) how ASMR content has evolved over time; and (v) how videos form exploratory clusters when represented in a joint feature space across the full observation period.

2 Method

All data collection and analysis code used in this study is available as supplementary material (see section 5). We collect ASMR-related video data from YouTube (`https://www.youtube.com`) using a keyword-based pipeline that combines the official YouTube Data API v3 (`https://developers.google.com/youtube/v3`) with an additional retrieval workflow implemented in the *pytubefix* library (`https://pytubefix.readthedocs.io`). The analysis is conducted on publicly available, video-level information (e.g. titles, descriptions, and engagement statistics) and does not reproduce or redistribute



Fig 3. Word cloud generated from the combined titles and descriptions of all 89,241 ASMR videos after custom stopword removal and text cleaning.

audiovisual content; where examples are discussed, we cite the original YouTube URLs. According to YouTube's guidance on copyright and fair use, such uses may be permissible for research purposes, noting that fair use determinations are context-specific and that applicable copyright exceptions vary between jurisdictions [1]. During data collection, we use a single English query keyword, "ASMR", which we pass identically to both the YouTube Data API search endpoint and the *pytubefix* search interface; in both cases, we restrict the results to standard videos (excluding non-video items such as channels and playlists).

To increase coverage beyond what the API alone returns and to obtain a richer set of metadata, the pipeline comprises two discovery branches. In the API-based branch, we use the YouTube Data API to perform paginated searches for the query "ASMR", restricted to standard videos. Each search retrieves up to 50 results per page and up to 100 pages (up to 5,000 candidates per search). To mitigate this per-search cap while covering the full history of ASMR content, we partition the study period into consecutive three-month upload windows and run separate API searches for each window, starting on 1 January 2008 and ending on 31 July 2026. In the scraping-based branch, we use *pytubefix* to issue the same keyword query against the public YouTube search interface, again restricted to videos and using YouTube's default ranking. For each video identifier discovered by either branch, we request the corresponding watch page using *pytubefix* and extract extended metadata, including title, description, duration in seconds, view and like counts, channel identifier, author name, upload date, and an initial language estimate based on the combined title–description text. When both API- and scraping-based metadata are available for the same video, we merge them into a single record; where fields are missing, we fall back to whichever source provides the information. If neither source provides a language label, we apply automatic language detection over the concatenated title and description using an off-the-shelf language-identification package. Before analysis, the JSON metadata enrichment step was configured to retrieve missing view and like counts while retaining existing non-missing values. Counts that could not be retrieved remained missing. Thus, the final recomputation used the available stored statistics rather than refreshing every video's view and like counts to a single common observation time. The final metadata were processed on 1 August 2026. The deposited JSON snapshot does not retain the optional per-record `metadataCollectedAt` and `languageSource`

[1] `https://support.google.com/youtube/answer/9783148?hl=en&sjid=15816069292466875886-EU`

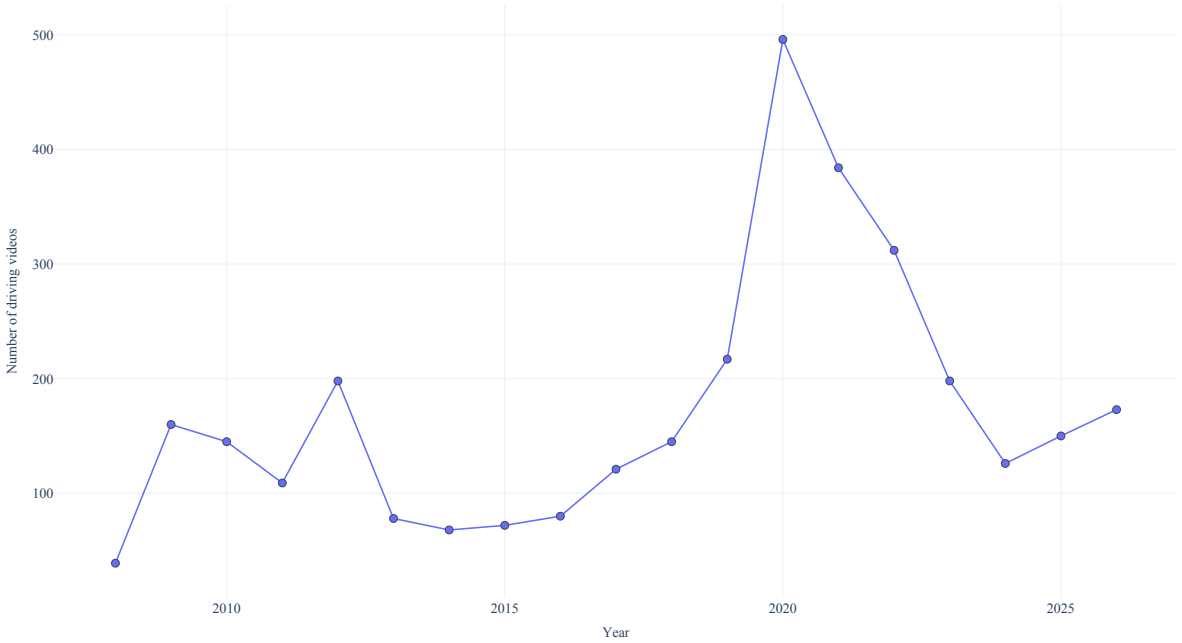


Fig 4. Yearly count of "driving"-themed ASMR videos, where the driving theme is defined via titles or descriptions containing the term *drive* or related phrases (e.g. "driving", "drive with me", "car", "road trip").

provenance fields. The reproducibility manifest therefore reports zero record-level metadata timestamps and an unknown language source for all records, while the stored language labels remain available for analysis. The exact input snapshot is identified by the SHA–256 digest `e260339147d2af21c7fc796d73f26e2f382cc29385443bcee65d189c34eccb18`.

All discovered videos are subjected to a uniform set of inclusion criteria. First, to enforce topical relevance, we require that the lowercase query keyword "asmr" appear in the video title (case-insensitive substring match); videos whose titles do not contain this token are discarded, even if they were retrieved by the API or the scraper. Second, to exclude YouTube Shorts and extremely brief clips, we require an estimated duration of approximately one minute, operationalised as at least 59 seconds. Videos below this threshold are removed when valid duration metadata are available; records that remain with missing or non-positive duration after enrichment are retained in an *unknown* duration bucket for transparent reporting and are excluded from duration-positive plots. After filtering, videos discovered through the two branches are deduplicated by video identifier, resulting in a unique set of ASMR-related videos for subsequent analysis.

For each video that passed these inclusion criteria, we constructed a structured record with the following per-video fields:

1. **Identifiers and channel metadata:** the unique video identifier, the associated channel identifier, and the channel's display name or author field.

2. **Textual fields:** the video title and description, as returned by YouTube at the time of collection. In all subsequent text-based analyses, we treat the concatenation of title and description as a single document.

3. **Temporal information:** the upload timestamp in UTC, the derived calendar date, and the upload year and month. For each video we also compute the number of days since upload relative to the fixed reference date of 1 August 2026, used in growth-related measures. Records with a missing or invalid upload date, and any records dated after this reference date, have missing growth rates and are excluded from views-per-day summaries.

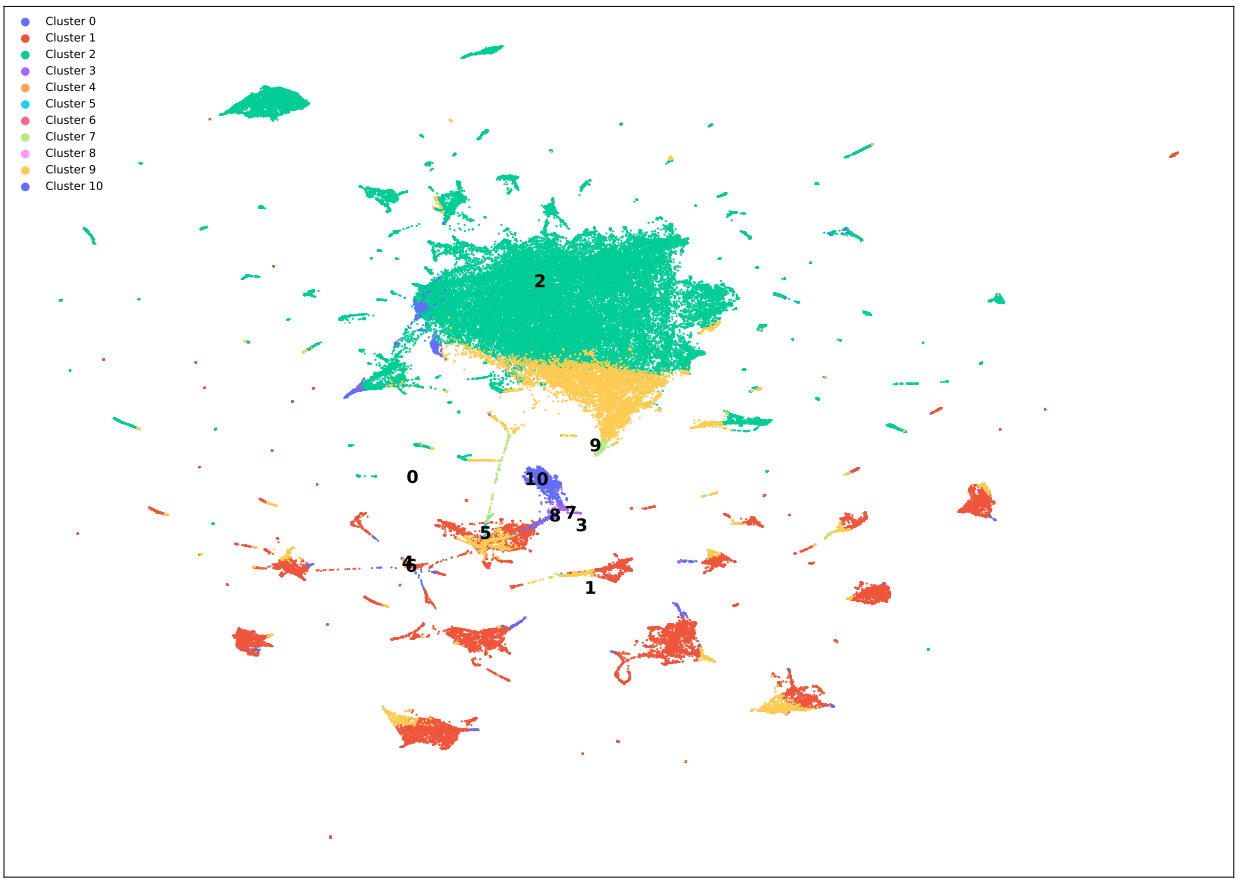


Fig 5. Two-dimensional UMAP projection of all 89,241 videos in a joint feature space combining text (TF–IDF over titles and descriptions), duration, engagement rate, views per day, and language. Each point is a video, coloured by its k-means cluster assignment.

4. **Duration:** the video duration in seconds and minutes, obtained by parsing ISO 8601 duration (`https://www.iso.org/iso-8601-date-and-time-format.html`) strings and/or watch-page metadata. We further discretise duration into coarse buckets with five levels: under 10 min, 10–30 min, 30–60 min, 60–180 min, and over 180 min, plus an *unknown* category for rare cases with missing values.

5. **Engagement statistics:** the total number of views and likes at the time of collection. From these we derive (i) the number of views per day since upload,

$$\text{views per day}(v) \;=\; \frac{\text{views}_v}{\text{days since upload}_v}\,, \tag{1}$$

and (ii) a per-video engagement rate defined as the ratio of likes to views whenever view counts are strictly positive,

$$\text{engagement}(v) \;=\; \frac{\text{likes}_v}{\text{views}_v}\,. \tag{2}$$

Videos with missing like counts or zero or missing view counts are assigned a missing engagement value. For any subset of videos S, let $S_+ = \{v \in S : \text{engagement}(v) \text{ is observed}\}$. We report the arithmetic mean across videos with observed engagement,

$$\text{mean engagement}(S) \;=\; \frac{1}{|S_+|}\sum_{v\in S_+} \text{engagement}(v)\,, \tag{3}$$

and refer to this quantity as *mean engagement*. As auxiliary context, we also obtain channel-level statistics from the YouTube Data API (total view count and total number of uploaded videos per channel) and compute an average number of views per uploaded video; this channel-average statistic is used to form a relative-views measure for some descriptive analyses but is not a primary focus of the present study.

6. **Language:** a normalised language code inferred from a combination of platform metadata (default audio or interface language) and automatic language detection on the concatenated title–description text, performed using the *langdetect* Python package (`https://pypi.org/project/langdetect/`). Where platform metadata and automatic detection disagree, we manually normalised obvious aliases (e.g. different codes for English) and treated the remainder as distinct categories.

7. **Title style features:** automatically derived indicators that characterise title formatting. These include the number of words and characters in the title and binary flags for stylistic devices: presence of brackets or parentheses, all-caps words of length at least three, exclamation marks, question marks, hashtags, and explicit "no talking" tags (e.g. "no talking", "no-talk"). These features are used in analyses relating title style to views and engagement.

8. **Content themes:** ten Boolean indicators capturing broad ASMR themes derived from cleaned, lowercased title–description text: whisper-focused content, no-talking or speech-free content, sleep-related content, binaural or 3D audio, role-play scenarios, ear-focused treatments, eating and mukbang-style content, keyboard and typing sounds, visually emphasised triggers, and driving-related content. Each indicator is set to true if the video's text matches a rule-based pattern for that theme and false otherwise.

9. **Growth category:** a categorical label that discretises the views-per-day measure into four levels: *slow*, *medium*, and *fast* for increasing ranges of views per day, and *unknown* for missing or non-positive values.

3 Results

For text-based analyses, word clouds were generated from titles, descriptions, and their concatenation for all 89,241 videos. The updated run produced 200-word clouds for each text source. Title frequencies are shown in Figure 1, and combined title and description frequencies are shown in Figure 3. Verb-only and spaCy keyword plots were skipped because the required spaCy language model was unavailable in the execution environment; consequently, the descriptive text analyses reported here rely on cleaned title and description text rather than on a new spaCy lemma-frequency table. Among records with available like statistics, the mean number of likes was 15,175.20 (SD 113,669.46), and the mean number of likes per day was 23.82 (SD 193.46) among records with the valid inputs required for that rate.

Automatic theme detection was performed using the prespecified rule-based pipeline on the lowercased title–description text. Because forced recomputation was enabled, the analysis recalculated all theme labels rather than reusing cached values. Theme-specific expressions and surface patterns were matched for whisper-focused content, no-talking content, sleep-related content, binaural or 3D audio, role-play scenarios, ear-focused treatments, mukbang or eating, keyboard or typing, visual triggers, and driving-related content. These variables are treated as textual theme-label indicators rather than definitive content categories: they record whether a video title or description contains matching labels or phrases, not whether the audiovisual content itself was manually verified. No manual precision estimate is reported for the present run, so all theme-level findings are interpreted descriptively and with this limitation in mind. This procedure yielded ten non-zero theme labels. The number and proportion of videos with each theme were: whisper (n=13,827; 15.49%), no talking (n=5,516; 6.18%), sleep related (n=15,725; 17.62%), binaural or 3D audio (n=5,612; 6.29%), role play (n=12,570; 14.09%), ear cleaning or ear focused (n=2,182; 2.45%), mukbang or eating (n=6,315; 7.08%), keyboard or typing (n=1,810; 2.03%), visual or hand movement triggers (n=6,111; 6.85%), and driving-related content (n=3,271; 3.67%). Growth categories were derived using fixed thresholds on views per day: videos with fewer than 100 views/day were labelled slow growth, those with 100–5,000

Table 1. Top ten language categories by number of videos. Values are means with SD in brackets for views, views/day, likes, and likes/day; engagement is the mean percentage. Language categories represented by at least 20 videos are provided in Table 5.

Language	n	Views	Views/day	Likes	Likes/day	Engagement ($\times 10^{-2}$)
English	**65087**	812,548 (5,861,636)	1,025.21 (10,368.97)	14,464 (117,986)	22.79 (198.71)	3.29
Korean	2961	2,288,151 (8,251,059)	1,757.90 (5,490.15)	26,046 (68,157)	20.57 (43.89)	2.59
Spanish	2656	521,065 (1,938,119)	768.45 (4,089.62)	15,026 (70,506)	22.72 (80.61)	4.90
Japanese	2595	1,233,464 (4,691,798)	1,886.17 (6,404.08)	19,868 (68,661)	35.18 (127.91)	1.97
Portuguese	1756	1,242,833 (5,769,754)	3,650.94 (38,172.61)	34,547 (127,371)	75.47 (419.96)	6.86
German	1445	619,812 (4,910,355)	771.63 (5,724.29)	10,852 (102,063)	17.13 (142.75)	3.97
French	1437	371,560 (4,963,130)	827.81 (15,391.94)	7,335 (67,211)	16.54 (205.19)	3.59
Dutch	1371	88,153 (731,312)	92.20 (1,224.08)	2,225 (23,612)	2.75 (39.83)	2.49
Indonesian	1286	1,054,398 (6,082,600)	2,527.47 (13,835.72)	15,550 (95,801)	42.71 (220.39)	6.11
Russian	1279	479,979 (7,009,948)	350.84 (4,375.50)	11,294 (156,383)	8.16 (97.13)	3.21

Table 2. Summary statistics for ASMR videos by title length bucket.

Title length bucket	n	Views (SD)	Views/day (SD)	Likes (SD)	Likes/day (SD)	Engagement ($\times 10^{-2}$)
$\leq$ 5 words	16121	472,801 (2,881,842)	659.42 (4,768.91)	10,125 (70,314)	16.66 (109.48)	4.03
6 to 10 words	40024	673,913 (5,175,637)	958.28 (11,024.84)	14,068 (123,474)	23.19 (193.81)	3.32
11 to 20 words	32856	1,274,042 (7,963,569)	1,533.35 (13,360.07)	18,535 (117,619)	27.84 (222.29)	3.44
$>$ 20 words	240	5,312,096 (16,205,670)	3,384.08 (9,826.28)	57,460 (142,618)	33.92 (75.70)	3.62

views/day medium growth, and those with at least 5,000 views/day fast growth. Videos with zero or missing values were assigned an unknown category.

Videos were stratified into duration buckets. Videos under 10 min accounted for 30,916 videos (34.64%), videos between 10 and 30 min for 38,190 videos (42.79%), videos between 30 and 60 min for 13,283 videos (14.88%), videos between 60 and 180 min for 4,918 videos (5.51%), and videos exceeding 180 min for 1,355 videos (1.52%); 579 videos (0.65%) had an unknown or non-positive duration. Mean views and engagement rates for each duration bucket are summarised in Table 3. For example, videos longer than 180 min averaged 1,381,584 views with a mean engagement rate of 3.09%, whereas videos under 10 min averaged 1,078,909 views with a mean engagement rate of 4.19%. Language-level summaries appear in Table 1, and title-length statistics appear in Table 2.

The relationship between duration and popularity was examined for the subset of 88,482 videos with positive duration and non-zero views (Figure 2). Duration ranged from 59 s to 244,211 s (median = 915 s, IQR = 370 to 1,662 s), with a mean of 1,526.57 s (SD = 3,300.07). Views ranged from 1 to 3.40×10^8 (median = 23,331.50; IQR = 1,406.25 to 250,884.50), with a mean of 873,593.20 (SD = 6,157,704.00). For the 88,485 videos with positive views, $\log_{10}(\text{views})$ had a mean of 4.25 and a standard deviation of 1.48. The D'Agostino–Pearson normality test yielded $k^2 = 4221.23$ and $p < 1.00 \times 10^{-3}$. The Shapiro–Wilk test yielded $W = 0.99$ and $p = 5.48 \times 10^{-22}$.

The descriptive statistics of views per day for the focal themes are summarised in Table 4. Among videos with non-missing views-per-day values, no-talking videos (n=5,512) had a mean of 1,236.20 views/day, sleep-related videos (n=15,701) had 798.09 views/day, whisper-labelled videos (n=13,824) had 394.63 views/day, binaural or 3D-audio videos (n=5,610) had 321.73 views/day, and driving-related videos (n=3,270) had 314.33 views/day. These values describe associations between creator-supplied textual labels and observed growth; they do not estimate causal effects of the themes.

Theme trends were computed for all years with valid upload dates (2008 to 2026). Overall trend tables were produced for all ten theme columns. The total themed-video counts in these trend tables were: whisper (13,825), no talking (5,514), sleep related (15,722), binaural or 3D audio (5,611), role play (12,565), ear cleaning or ear focused (2,181), mukbang or eating (6,315), keyboard or typing (1,810), visual triggers (6,111), and driving-related content (3,271). By-language trend tables contained 739 language–year rows for each theme, but no language met the minimum threshold of 30 year groups for by-language trend plotting. A yearly trend for driving-related content is shown in Figure 4.

Table 3. Summary statistics by duration bucket.

Duration bucket	n	Views (SD)	Views/day (SD)	Likes (SD)	Likes/day (SD)	Engagement ($\times 10^{-2}$)
Under 10 min	30916	1,078,909 (8,157,855)	2,075.55 (18,457.52)	24,874 (190,498)	49.81 (336.90)	4.19 (6.86)
10 to 30 min	38190	790,838 (5,518,309)	585.11 (3,506.68)	10,950 (51,834)	10.56 (43.83)	3.18 (3.99)
30 to 60 min	13283	611,647 (2,090,947)	563.98 (1,620.30)	9,455 (27,384)	13.50 (47.76)	2.65 (2.47)
60 to 180 min	4918	789,979 (2,663,295)	825.11 (2,451.35)	11,029 (28,858)	15.42 (36.23)	4.58 (11.72)
Over 180 min	1355	1,381,584 (5,847,939)	1,293.15 (3,728.16)	15,384 (61,844)	18.30 (43.75)	3.09 (4.44)
Unknown	579	7,547,073 (12,283,370)	13,060.09 (20,111.47)	23,114 (31,162)	54.46 (44.97)	1.34 (1.23)

Table 4. Summary statistics for videos containing each thematic category (theme present only). n_{theme} is the number of videos flagged with the theme. n_{vpd} is the subset with non-missing views/day used for views/day and likes/day.

Theme	n_{theme}	n_{vpd}	Views (SD)	Views/day (SD)	Likes (SD)	Likes/day (SD)	Engagement ($\times 10^{-2}$)
Binaural / 3D audio	5612	5610	615,943 (2,435,833)	321.73 (1,132.38)	8,349 (30,206)	6.70 (29.39)	2.21 (2.26)
Driving	3271	3270	321,066 (3,495,719)	314.33 (3,313.88)	5,233 (35,841)	7.81 (54.07)	3.29 (5.87)
No talking	5516	5512	1,679,134 (9,229,991)	1,236.20 (9,388.24)	20,713 (78,356)	18.32 (135.35)	3.43 (7.30)
Sleep related	15725	15701	707,544 (3,648,560)	798.09 (6,051.61)	11,365 (69,552)	17.74 (101.14)	3.05 (3.42)
Whisper	13827	13824	469,061 (2,390,509)	394.63 (1,857.82)	7,337 (31,085)	10.44 (53.91)	2.64 (2.16)

Finally, we visualise the 11-cluster solution using UMAP (Figure 5). K-means clustering was fit in a joint feature space combining text, duration, engagement rate, views per day, and language. The full UMAP embedding contains valid two-dimensional coordinates for all 89,241 videos. Cluster sizes range from 12 to 55,264 videos. The largest cluster contains 55,264 English-language videos and has mean growth of 516.28 views/day. A second large, multilingual cluster contains 20,426 videos and has mean growth of 760.90 views/day. Other clusters include 2,054 videos averaging 166.81 minutes and 1,248.83 views/day, 448 videos averaging 568.72 minutes and 816.96 views/day, and 19 videos averaging 1,615.78 minutes and 1,137.25 views/day. Two small low-growth clusters contain 78 and 655 videos with means of 2.01 and 2.16 views/day, respectively. Three high-growth outlier clusters contain 550, 91, and 12 videos and average 46,561.13, 188,211.54, and 660,146.26 views/day, respectively. The UMAP map should therefore be interpreted as an exploratory full-corpus visualisation of the clustering solution rather than as a definitive taxonomy of ASMR subgenres.

4 Discussion

This study's main contribution is an ecosystem-level, multilingual longitudinal map of explicitly labelled ASMR video production on YouTube, produced by combining large-scale metadata analysis with lexical description, rule-based theme-label indicators, and exploratory unsupervised clustering. Whereas much prior research has focused on individual creators, specific triggers, or comment-based ethnographies, our dataset captures 89,241 videos uploaded between 2008 and 2026 from 18,024 channels across 81 language categories. This scale and temporal range allow us to situate familiar ASMR formats within a broader and more heterogeneous media ecology and to describe how explicitly labelled ASMR has diversified by language, duration, textual theme label, and engagement over time.

Lexical analysis (Figure 1 and Figure 3) indicates that ASMR titles and descriptions are organised around a stable vocabulary of relaxation, sensory experience, and affective orientation. In the updated execution, word clouds were generated successfully from titles, descriptions, and their concatenation for all 89,241 videos, whereas spaCy keyword and verb-lemma tables could not be populated because the required language model was unavailable. We therefore treat the lexical evidence as descriptive rather than as a formal lemma-frequency model. These word-cloud patterns show how creators label and frame ASMR videos, and they echo experimental and ethnographic accounts that discuss relaxation, sleep support, comfort, care, and interpersonal address [1, 7–9, 19, 22, 24]. They should not be read as evidence that the videos in this corpus caused relaxation, improved sleep, or produced physiological effects.

The refreshed theme indicators also revise the interpretation of driving-related ASMR. In the final run, driving-related content accounts for 3,271 videos, or 3.67% of the 89,241-video corpus. Its temporal profile (Figure 4) therefore indicates a visible but comparatively small textual label category, not a dominant format. One plausible interpretation is that some creators use car, driving, road-trip, or ambient mobility labels to frame continuous soundscapes and everyday infrastructures as ASMR settings. However, because this indicator is rule based and text only, we treat driving-related ASMR as an emergent labelled niche whose boundaries require manual or multimodal validation rather than as a settled content category.

Language-level results (Table 1) underscore the globalisation of ASMR production. English remains dominant (72.93%), but substantial activity is evident in Korean, Spanish, Japanese, Portuguese, German, French, Dutch, Indonesian, Russian, and other language communities. Among categories with at least 20 videos, the raw code *fr-fr* has the highest mean views per day (25,916.35; n=20), while the undetermined code *und* has the highest mean engagement (13.70%; n=21). These maxima are based on small categories and should be interpreted cautiously. Together, the broader distribution suggests that ASMR is not merely replicated across languages but is actively reshaped within local creative cultures, consistent with arguments that ASMR practices are culturally situated rather than uniform across linguistic communities [24, 25]. The present findings therefore support a shift from English-centric descriptions towards comparative work that attends to platform-native performance metrics across linguistic communities.

Duration-based patterns point to a mixed ecology of ASMR formats. Videos under 10 minutes account for a substantial share of production (34.64%) and show the highest mean views per day among the defined duration buckets, consistent with quick-access trigger content embedded in everyday routines. Very long videos exceeding 180 minutes remain rare (1.52%) but still accumulate high mean total views, consistent with uses as background ambience and sleep support. Across the corpus, the heavy-tailed distribution of views and strong deviations from normality in $\log_{10}(\text{views})$ confirm that ASMR, like other YouTube genres [31], exhibits pronounced inequality: a small number of highly successful videos capture disproportionate attention, while the majority attract modest audiences.

Theme-based indicators further nuance common assumptions about which formats "perform best." When views/day is computed over videos with valid growth values, no-talking videos (n=5,512; 1,236.20 views/day) and sleep-related videos (n=15,701; 798.09 views/day) show higher mean growth than whisper videos (n=13,824; 394.63 views/day). Binaural or 3D-audio content remains comparatively low growth (n=5,610; 321.73 views/day), as does driving-related content (n=3,270; 314.33 views/day). These patterns suggest that contemporary visibility is not driven solely by canonical whisper-based presentation or by technically sophisticated audio setups. Instead, growth appears to reflect an interplay of creative choices, audience use cases, and recommender dynamics, with particular advantage for formats that reduce linguistic dependence (no talking) or supply explicit sleep support.

Clustering analysis provides an exploratory lens on ASMR's internal heterogeneity. Using an eleven-cluster k-means solution in a joint feature space combining text, behavioural metrics, and language, we describe groups that differ markedly in typical duration, language composition, views per day, and mean engagement. A very large English cluster (55,264 videos) shows moderate mean growth (516.28 views/day), consistent with a broad, general-purpose ASMR core. A second large multilingual cluster (20,426 videos) shows higher mean growth (760.90 views/day), indicating a substantial non-English production regime that is not simply peripheral to the English core. Additional clusters capture distinct format logics, including long-duration groups averaging 166.81, 568.72, and 1,615.78 minutes. The clustering results also reveal very low-growth clusters with minimal mean views/day alongside several high-growth outlier clusters. The coexistence of extreme high- and low-growth clusters reinforces that engagement metrics can be unstable in sparse or low-view regimes and that growth and engagement should be interpreted jointly rather than in isolation.

Finally, the UMAP visualisation of the clustering solution suggests multiple dense regions separated by low-density transition zones, but this should be interpreted as an exploratory projection of the fitted feature space rather than as a definitive map of ASMR subgenres. Dense regions plausibly correspond to widely shared production formulae, whereas smaller clusters and isolated points reflect unusual combinations of themes, languages, and presentation styles, including niche high-growth configurations. The updated UMAP map contains two-dimensional coordinates for every video in the 89,241-video corpus; only optional browser-oriented interactive outputs are downsampled for responsiveness. Overall, the results portray

explicitly labelled YouTube ASMR as a multilingual and longitudinally diversifying field, structured by overlapping stylistic and functional labels, duration regimes, and engagement patterns, while leaving causal claims about viewer experience and definitive subgenre boundaries for future validated research.

4.1 Limitations and future work

This study has several important limitations that also suggest concrete directions for future research. Our corpus is restricted to YouTube videos of approximately one minute or longer, operationalised as at least 59 seconds, and retrieved via a keyword based pipeline centred on the query "ASMR". In practice, this means that we focus on ASMR as it is explicitly labelled by creators and surfaced through keyword search, rather than on the full universe of ASMR adjacent or implicitly soothing content. This constraint is especially salient given the scale of the contemporary corpus and the increased prominence of shorter, clip like, and hybrid formats that may not use the label consistently. Short form variants of soothing or ASMR like content have become more visible on TikTok, Instagram Reels, Facebook, and YouTube Shorts, but such materials fall largely outside our sampling frame. The present findings therefore characterise patterns in explicitly labelled YouTube ASMR rather than the full cross platform ecosystem. Future work could extend the corpus by combining richer query sets (for example, sleep, no talking, whisper, binaural, roleplay, tapping, or trigger related terms) with short form data from multiple platforms and explicitly compare stylistic patterns, trigger types, and engagement dynamics across duration regimes and ecosystems.

A further limitation is that our analysis is entirely based on platform metadata and textual information (titles, descriptions, and related fields). The rule-based theme variables were not manually validated in the present run; they should therefore be interpreted as scalable text-label indicators rather than as high-confidence annotations of the audiovisual content. Such metadata is creator dependent, often incomplete, and not standardised, which can lead to under detection or misclassification of ASMR themes. Many core ASMR triggers, especially auditory triggers such as tapping, scratching, mouth sounds, or brushing, cannot be reliably inferred from text alone, and nuanced differences in performance style (for example, microphone technique, pacing, or camera work) remain opaque at the metadata level. This limitation is amplified by the expanded multilingual coverage, because theme keywords and idiomatic trigger descriptions vary across languages and communities. Subsequent studies should incorporate multimodal features, including audio and video based descriptors (for example, spectrogram features, spatial audio cues, visual trigger detection, gesture and camera movement analysis), to obtain a more faithful representation of the sensory content of ASMR videos and to reduce reliance on language specific surface patterns.

Because the study does not observe viewers, comments in depth, experimental outcomes, sleep, wellbeing, or physiology, it cannot determine whether labelled ASMR videos produce the experiences or effects discussed in prior literature. The focus on a single platform also introduces platform specific biases. YouTube has its own recommendation algorithms, audience composition, and production norms, which shape how ASMR content is produced, surfaced, and engaged with. As a result, the engagement metrics, language distributions, and temporal trends observed here may not generalise to other platforms where ASMR culture, audience behaviour, and content curation differ. Moreover, growth metrics such as views per day can reflect both audience demand and platform distribution effects, which may vary substantially across platforms and between long form and short form regimes. A cross platform perspective that combines data from ecosystems such as TikTok, Instagram, and Facebook would allow researchers to examine how platform design, affordances, and recommendation logic influence ASMR production and consumption, and to test whether the length and theme patterns observed here extend beyond YouTube.

Methodologically, the use of UMAP as a non linear embedding for visualising similarities among videos comes with important constraints: it is sensitive to hyperparameters, preprocessing choices, and random seeds, and the resulting map is still an exploratory projection rather than a definitive taxonomy. The full-corpus UMAP addresses the practical limitation that the earlier t-SNE workflow became difficult to compute and render at this scale, but it does not remove the need for careful interpretation of distances and cluster boundaries. This caution is particularly important given the extreme cluster size imbalance in the updated dataset, where a small number of very large clusters coexist with tiny outlier clusters that can be visually emphasised. Future work could explore more interpretable and reproducible embedding strategies, such as transformer based text embeddings, multimodal audio visual representations, or supervised

embeddings aligned with manually annotated ASMR categories, and then revisit clustering or community detection using these representations.

Finally, even though our sampling strategy spans the period from January 2008 to July 2026 and partitions searches into three month upload windows, the temporal and linguistic coverage of the dataset remains uneven. Some years and some languages are sparsely represented, and keyword plus date bounded retrieval cannot guarantee complete recall for high volume periods. In addition, views-per-day measures use the fixed reference date of 1 August 2026. Records without a valid upload date or growth value are excluded from analyses using views per day, while the most recent uploads have short exposure intervals and may therefore yield comparatively unstable rates. The deposited snapshot also lacks record-level provenance for the source of each stored language label and the time at which each individual metadata record was retrieved. Because existing non-missing view and like counts were retained during enrichment, the engagement statistics do not represent a synchronised observation of all videos at one date and may partly reflect differences in collection timing. This imbalance can make longitudinal interpretations less stable for early years, recent uploads, or low-sample languages and can inflate the apparent importance of heavily represented periods. Future studies should explicitly model such imbalances, for example, using hierarchical or time series models that account for varying data density, constructing more balanced sampling schemes across time, language, and platform, or combining keyword based retrieval with channel centric sampling. Together, these extensions, short form and cross platform data, multimodal content analysis, more robust embeddings, and temporally aware modelling, would provide a more comprehensive picture of how ASMR is produced, experienced, and transformed across the contemporary media landscape.

Across 89,241 explicitly labelled ASMR videos uploaded from 2008 to 2026, YouTube ASMR appears as a large, multilingual, and internally heterogeneous media ecosystem. English accounts for most videos, but substantial production occurs across many other language categories. Duration and creator-supplied theme labels are associated with markedly different growth profiles: videos under 10 minutes have the highest mean views per day among the defined duration groups, while no-talking and sleep-related labels show higher mean growth than whisper, binaural, or driving-related labels. The clustering and UMAP results further show that this corpus cannot be reduced to a single dominant format, although the resulting groups remain exploratory rather than definitive subgenres. These findings establish a reproducible descriptive baseline for future manually validated, multimodal, cross-platform research while avoiding causal claims about viewer experience or wellbeing.

5 Supplementary material

In line with current open science practices and recommendations for transparency in user research [32], the authors openly provide these research artefacts to support reproducibility, collaboration and further advancements in the field. The full dataset collected during the study, together with all analysis and visualisation code, is available at `https://www.dropbox.com/scl/fo/7fyi4syvs059bctagj8ix/AOyIQjQrfQBCGgTpqONHQC4?rlkey=ro13uemf57ownq2gmparfoxw8&st=tj5jlorh&dl=0`. The supplementary package includes a reproducibility manifest recording the dataset SHA–256 digest, the fixed analysis reference date of 1 August 2026, the random seed, theme-rule version, cluster count, Python version, and package versions used for the final analysis.

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Table 5. Language-level statistics for categories represented by at least 20 videos. Values are means with SD in brackets for views, views/day, likes, and likes/day; engagement is the mean percentage. Raw language codes are retained when no normalised display name was available.

Language	n	Views	Views/day	Likes	Likes/day	Engagement ($\times 10^{-2}$)
English	**65087**	812,548 (5,861,636)	1,025.21 (10,368.97)	14,464 (117,986)	22.79 (198.71)	3.29
Korean	2961	2,288,151 (8,251,059)	1,757.90 (5,490.15)	26,046 (68,157)	20.57 (43.89)	2.59
Spanish	2656	521,065 (1,938,119)	768.45 (4,089.62)	15,026 (70,506)	22.72 (80.61)	4.90
Japanese	2595	1,233,464 (4,691,798)	1,886.17 (6,404.08)	19,868 (68,661)	35.18 (127.91)	1.97
Portuguese	1756	1,242,833 (5,769,754)	3,650.94 (38,172.61)	34,547 (127,371)	75.47 (419.96)	6.86
German	1445	619,812 (4,910,355)	771.63 (5,724.29)	10,852 (102,063)	17.13 (142.75)	3.97
French	1437	371,560 (4,963,130)	827.81 (15,391.94)	7,335 (67,211)	16.54 (205.19)	3.59
Dutch	1371	88,153 (731,312)	92.20 (1,224.08)	2,225 (23,612)	2.75 (39.83)	2.49
Indonesian	1286	1,054,398 (6,082,600)	2,527.47 (13,835.72)	15,550 (95,801)	42.71 (220.39)	6.11
Russian	1279	479,979 (7,009,948)	350.84 (4,375.50)	11,294 (156,383)	8.16 (97.13)	3.21
Filipino	907	783,167 (7,315,670)	1,157.30 (10,711.26)	10,427 (84,832)	22.08 (202.43)	4.74
Italian	779	360,652 (2,635,606)	561.18 (4,208.10)	9,712 (76,071)	15.37 (112.09)	4.34
Estonian	579	1,503,424 (12,665,107)	909.91 (5,748.30)	13,785 (65,236)	11.47 (46.84)	6.06
Unknown	575	NA	NA	NA	NA	NA
Turkish	502	272,278 (1,116,827)	270.96 (773.00)	4,853 (23,456)	5.57 (14.99)	3.51
Vietnamese	364	9,044,790 (29,093,350)	4,981.06 (14,700.33)	84,545 (236,556)	44.89 (115.13)	2.00
Norwegian	252	269,020 (1,618,906)	941.30 (9,244.38)	7,392 (38,516)	24.89 (181.51)	4.63
Danish	245	634,136 (3,735,185)	1,791.54 (8,006.11)	15,572 (97,535)	44.10 (161.50)	4.56
Thai	237	510,833 (1,709,990)	365.89 (1,427.86)	8,998 (26,268)	7.07 (27.14)	4.13
Swedish	225	1,013,430 (13,156,417)	2,019.31 (25,269.12)	18,178 (222,555)	35.79 (426.85)	3.84
Catalan	219	491,508 (1,384,854)	1,663.98 (6,176.86)	13,325 (34,251)	43.16 (115.21)	4.02
Arabic	211	485,148 (1,644,859)	1,181.90 (2,110.81)	13,688 (46,638)	36.27 (66.23)	4.55
Afrikaans	183	238,790 (1,400,050)	379.41 (1,130.33)	4,765 (19,193)	15.21 (49.87)	5.06
so	172	366,226 (3,325,625)	298.72 (2,215.62)	4,996 (26,620)	7.14 (23.27)	6.15
Polish	169	181,376 (459,652)	106.98 (241.19)	4,157 (13,791)	3.05 (9.20)	3.21
Portuguese (Portugal)	143	637,432 (2,818,678)	2,376.37 (9,440.55)	16,721 (74,539)	53.07 (208.43)	6.96
Romanian	143	879,421 (6,083,567)	1,429.74 (8,229.08)	21,777 (163,850)	41.12 (215.39)	4.94
Bulgarian	125	1,325,307 (12,261,840)	985.59 (7,335.44)	47,763 (469,842)	32.12 (279.11)	2.92
Spanish (Mexico)	122	136,319 (993,258)	185.23 (608.26)	3,859 (22,274)	7.20 (19.90)	4.10
Swahili	110	71,777 (492,598)	185.29 (1,244.06)	6,230 (21,580)	16.33 (55.03)	5.35
zxx	83	1,522,266 (7,678,979)	628.74 (2,433.08)	8,560 (27,260)	6.47 (22.51)	2.66
en-ca	79	103,290 (312,177)	187.10 (769.06)	2,585 (7,319)	7.62 (36.62)	4.95
Hungarian	73	95,592 (602,665)	81.32 (317.15)	3,402 (15,612)	7.50 (31.82)	4.55
lt	69	238,097 (614,976)	776.59 (2,731.48)	6,862 (19,254)	27.33 (102.76)	3.98
sl	68	474,670 (2,283,114)	568.28 (2,349.61)	12,841 (59,863)	15.58 (61.38)	6.42
Hindi	67	1,695,552 (11,730,627)	5,129.65 (34,642.42)	3,374 (20,701)	12.64 (72.20)	4.79
cy	66	84,457 (368,997)	242.66 (1,497.59)	2,156 (10,141)	5.62 (24.49)	6.33
Portuguese (Brazil)	59	654,154 (1,624,624)	3,354.64 (8,193.23)	18,622 (36,801)	96.15 (187.80)	5.90
es-419	59	215,816 (242,071)	731.17 (928.13)	7,635 (9,656)	27.45 (41.85)	4.19
Finnish	51	104,755 (187,362)	129.21 (358.71)	1,839 (3,067)	3.61 (14.23)	5.49
Chinese (Simplified)	47	620,099 (2,951,167)	505.48 (1,986.64)	7,096 (33,040)	7.05 (27.14)	1.78
sq	40	304,850 (1,513,495)	791.22 (3,278.80)	4,305 (16,620)	14.78 (42.28)	8.92
lv	36	171,832 (658,710)	455.93 (1,442.01)	7,637 (26,571)	20.08 (60.08)	9.45
hr	36	177,004 (821,097)	1,016.66 (5,835.49)	2,334 (6,091)	5.24 (19.36)	4.10
Chinese	31	32,270 (109,211)	15.02 (38.57)	116 (277)	0.07 (0.13)	1.90
Czech	30	83,232 (196,250)	118.05 (299.68)	2,498 (5,036)	5.68 (19.10)	6.07
mk	21	494,434 (1,463,160)	1,295.25 (3,941.77)	9,842 (19,591)	24.27 (66.01)	3.42
und	21	1,543 (3,713)	1.58 (5.12)	36 (117)	0.02 (0.06)	**13.70**
fr-fr	20	**12,003,420 (36,992,870)**	**25,916.35 (64,480.50)**	285,573 (882,335)	597.65 (1,534.40)	3.61